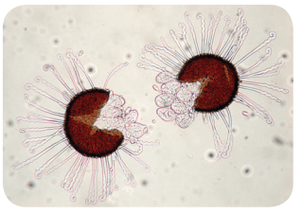
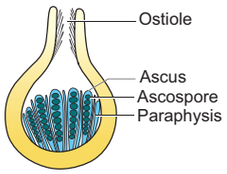


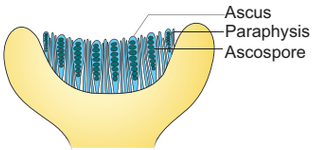
(**a**) *Peziza*



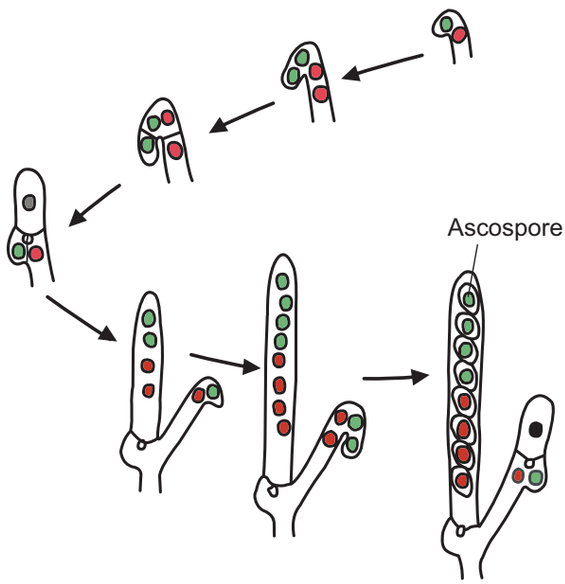
(**b**) Cleistothecium



(**c**) V.S. of Perithecium



(**d**) V.S. of Apothecium



(**e**) Steps involved in the development of Ascus

Figure 1.21: Structure and reproduction in Ascomycetes

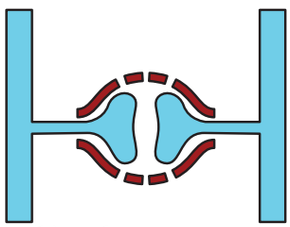
- Asexual reproduction takes place by fission, budding, oidia, conidia, and chlamydospore.
- Sexual reproduction takes place by the fusion of two compatible nuclei.
- Plasmogamy is not immediately followed by karyogamy, instead a dikaryotic condition is prolonged for several generations.
- A special hyphae called ascogenous hyphae is formed.
- A crozier is formed when the tip of the ascogenous hyphae recurves forming a hooked cell. The two nuclei in the penultimate cell of the hypha fuse to form a diploid nucleus. This cell forms young ascus.
- The diploid nucleus undergo meiotic division to produce four haploid nuclei, which further divide mitotically to form eight nuclei. The nucleus gets organised into 8 ascospores.
- The ascospores are found inside a bag like structure called ascus. Due to the presence of ascus, this group is popularly called "Sac fungi".
- Asci gets surrounded by sterile hyphae forming fruit body called ascocarp.
- There are 4 types of ascocarps namely **Cleistothecium** (Completely closed), **Perithecium** (Flask shaped with ostiole), **Apothecium** (Cup shaped, open type) and **Pseudothecium**.

Basidiomycetes

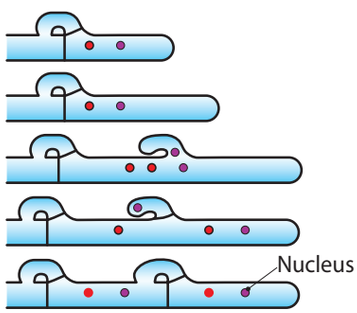
- Basidiomycetes include puff balls, toad stools, Bird's nest fungi, Bracket fungi, stink horns, rusts and smuts (Figure 1.22).



(**a**) *Geaster*



(**b**) Dolipore septum



(**c**) Clamp connection

Figure 1.22: Structure and Reproduction in Basidiomycetes

- The members are terrestrial and lead a saprophytic and parasitic mode of life.

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